
Healthy Soils Thriving Farms
Self Study Report
Ph.D. (Agril. Engg.) with specialization in Soil and Water Conservation Engineering
SOIL | WATER | LAND | PEOPLE | SUSTAINABLE TOMORROW
Conserve Water Secure Future
GIS & REMOTE SENSING FOR WATERSHED MANAGEMENT
SOIL AND WATER CONSERVATION STRUCTURES (CONTOUR BUNDS & TERRACES)
CHECK DAMS AND RAINWATER HARVESTING
SOIL PHYSICAL ANALYSIS LABORATORY
SOIL WATER LIFE
FOR A SUSTAINABLE AGRICULTURE
SOIL HYDRAULICS AND WATER MOVEMENT LABORATORY
RUNOFF AND SOIL EROSION STUDIES
SURVEYING AND LAND EVALUATION
IN-SITU SOIL MOISTURE MONITORING
SOIL SURVEY AND LAND RESOURCE ASSESSMENT
WATERSHED DEVELOPMENT AND WATER RESOURCE MANAGEMENT
ANGRAU
Acharya N. G. Ranga Agricultural University
Dr. N.T.R. College of Agricultural Engineering, Bapatla
ESTD 1963
BAPATLA
Conserve Soil... Harvest Water... Ensure Food Security...

Self Study Report 2026

Ph. D. Programme

Department of Soil and Water Conservation Engineering

Dr. N.T.R College of Agricultural Engineering, Bapatla

INDEX

S. No.	Self Study Report	Page No.
1	Ph. D. (Soil and Water Conservation Engineering)	4-48

ACCREDITATION REPORT – TASK COMMITTEE

Chairman : Dr D. D. Smith, Associate Dean Coordinator

: Dr G. Ravi Babu, Dept. of SWCE

Members : Dr K. Krupavathi, Associate Professor, Dept of IDE

Dr N. Ashok Kumar, Assistant Professor, Dept. of IDE Er

M. Karunya, Assistant Professor, Dept. of SWCE

Smt. V. Padma Tulasi, Administrative Officer





Contents

	PhD (Soil and Water Conservation Engineering)	
6.4.1	Brief History of the Degree Programme	**4**
6.4.2	Faculty Strength	**5**
6.4.3	Technical and Supporting Staff	**6**
6.4.4	Classrooms and Laboratories	**7**
6.4.5	Conduct of Practical and Hands-on Training	**13**
6.4.6	Supervision of Students in PG Programmes	**14**
6.4.7	Feedback of Stakeholders (Students, Parents, Industries, Employers, Farmers, etc.)	**16**
6.4.8	Student Intake and Attrition in the Programme for Last Five Years	**22**
6.4.9	ICT Application in Curricula Delivery	**23**
6.5.3	Learning resources	**38**
6.5.4	Student development	**40**
6.5.9	Outcome/Output	**42**
6.4.11	Certificate	**45**

6.4 SELF STUDY REPORT FOR ACCREDITATION OF PROGRAMME

Ph.D. (Agricultural Engineering - Soil and Water Engineering)
Dr N.T.R College of Agricultural Engineering, Bapatla

6.4.1. Brief History of the Degree Programme

B.Tech. (Agricultural Engineering) undergraduate (UG) degree programme was started in August, 1983 with an intake of 20 students in the Department of Agricultural Engineering, Agricultural College, Bapatla. From the next year onwards, the seats were increased to 30. The UG programme was upgraded as the College of Agricultural Engineering (CAE), Bapatla in 1990 based on the recommendations of a high-level committee appointed by the Government of Andhra Pradesh. It is the first college of agricultural engineering in Acharya N.G. Ranga Agricultural University. It was started with the aim of imparting knowledge and skills to the students in the disciplines of agricultural engineering education to serve the farming community and the rural people of the combined state of Andhra Pradesh and to make agriculture more remunerative to the farmers by bringing in the engineering interventions in agriculture. The College was shifted to its new premises in May, 1994. Since then, the College expanded rapidly both academically and physically. Many infrastructural facilities were added in terms of buildings and equipment for improving the academic status in agricultural engineering. This college was accredited by Indian Council of Agricultural Research, New Delhi during 1999-2000 in the first phase itself based on ICAR's Review Team Report and during the subsequent phases also. The latest accreditation of ICAR for this College was in 2021. The College was under the faculty of Agriculture until 2010 when a new Faculty of Agricultural Engineering and Technology was established by the University as per the provisions of the University Act. The College was renamed as Dr N.T.R. College of Agricultural Engineering (Dr N.T.R CAE) in the year 2018 in memory of Late Dr N.T. Rama Rao the then Chief Minister of combined Andhra Pradesh and who was the person laid foundation stone for the College building on 24-02-1988. Today, Dr N.T.R CAE is widely recognized for its quality education; need based research and transfer of technology in service of farming community.

Ph.D. (Agricultural Engineering) programme was initiated during the year 2012-13 in the Department of Soil and Water Engineering. Ph.D. students are supported with a monthly stipend of Rs.10,000/-. ICAR specified syllabus is being followed for teaching Ph.D. (Agricultural Engineering) course.

The College has state-of-the-art infrastructure and facilities and the academic environment which combines academic rigor with high level of interaction with government departments in serving the farming community of the nation in general and the present state of Andhra Pradesh after bifurcation in particular. The driving force behind the high level of academic excellence and quality education is the dedicated team of over thirty core faculty drawn from different specializations.

VISION

To make Dr N.T.R College of Agricultural Engineering, Bapatla an integrated center for academic excellence of agricultural engineering education known for its contribution to agriculture and allied fields in making agriculture more remunerative to farmers

MISSION

To serve the farming community and enhance their quality of life through excellence in education, research, development and transfer of agricultural technologies

6.4.2. Faculty Strength

The Ph.D. programme requires faculty from department of Soil and water conservation Engineering, Irrigation and drainage Engineering, Farm machinery and Power Engineering and other applied sciences. The faculty available for the programme is furnished below.

Soil and Water Conservation Engineering				
S No	**Faculty**	**Faculty Sanctioned**	**Faculty in place**	**Vacant position**
1.	Associate Dean	1	1	0
2.	Professor	1	1+3*	0
3.	Associate Professor	1	1+1*	0
4.	Assistant Professor	2	3+1*	0
	Total	**5**	**10**	**0**

(* indicates staff from Ag. College, Bapatla)

6.4.3. Technical and Supporting Staff

There is no separate Office Staff for running the Ph.D. (Agricultural Engineering) programme.

The Office Staff of the college available to support the B Tech (agri Engg) degree are also available for running the Ph.D. (Agricultural Engineering) programme.

Sanctioned and In-Position Laboratory Staff at Dr N.T.R CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Laboratory Staff Recommended by ICAR
1	Workshop Supervisor	1	1*	-	27 Workshop Staff, Laboratory Staff, Driver, Technicians
2	Technician (Mech. Gr.I)	3	5*	-	
3	Technician (Mech.Gr.II)	2	2*	-	
4	Draughtsman Grade III	1	1*	-	
5	Carpenter Grade II	1	1*	-	
6	Welder	1	1*	-	
7	Blacksmith	1	1*	-	
8	Turner	1	1*	-	
9	Driver	1	1+3*	-	
10	Workshop Helper	2	2*	-	
11	Luskar	2	2*	-	
12	Watch & Ward Staff	0	8*	-	
12	Laboratory Assistant	-	8*	-	14
13	Computer operator	-	3*	-	
	Total	**16**	**41**	**-**	**41**

*Positions filled with timescale, outsourcing, or contractual employees.

Sanctioned and In-Position Office Staff at Dr N.T.RCAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Office Staff Recommended by ICAR
1	Administrative Officer	1	1	-	1
2	Superintendent	1	1	-	20
3	Senior Assistant	2	2*	-	Clerks, steno,
4	Junior Office Assistant	2	1+7*	-	
5	Record Assistant	1	1	-	
6	Office Subordinate	2	2	-	
7	Office supporting staff	-	8*	-	-
8	Messenger/Peon		10*		8
9	Watchman	1			
	Total	**10**	**36**	**-**	**29**

** Staff recruited on Time scale/Outsourcing/Contract*

6.4.4. Classrooms and Laboratories

There is one dedicated classroom with good lighting and ventilation being used for M. Tech. (Agricultural Engineering) degree programme in the department of Soil and Water Engineering for conducting theory classes. The classroom is equipped with LCD projector, chalk board, furniture, etc. The department has different laboratories to conduct the practical classes regularly. Watershed model along with a farm pond is established in the College for students' experimentation. The department has dedicated laboratories. Department-wise distribution of laboratories is furnished here under:

A. Soil and Water Conservation Engineering

1. Soil and Water Conservation Laboratory
2. Watershed
3. Hydroponics
4. Surveying and Leveling Laboratory

B. Irrigation and Drainage Engineering

1. Hydraulics Laboratory
2. Irrigation Filed Laboratory
3. Micro Irrigation demo block
4. Wells and Pumps Laboratory

Sl. No.	Name of the laboratory	Name of the Important Equipment/Instruments
1	Field Irrigation Lab	Well irrigation system, Fields with drip irrigation system, Green houses, Fertigation equipment, Soil moisture measuring sensors, Water meter, Current meter, Mole plough, Power weeder, Rain gauges, Evaporimeter, Automatic weather station, Hydraulic flume, Soil augers, Hot air oven, etc.
2	Well and Pumps Laboratory	Cut models of different pumps, Different impellers, Pumps, Motors, Solar pumping system, etc,
3	Watershed	Watershed model, Farm pond, Soil and water conservation structures, etc.
4	Soil and Water Conservation Engineering Lab	Double ring infiltrometer, Soil moisture measuring devices, Parshall flumes, RBC flumes, Total station, GPS, Electronic theodolite, etc.
5	Hydroponics	A full-fledged hydroponics laboratory with all ative and passive equipment like linear and vertical NFT, DWC, Ebb and Flow system, Grow bag with drip,

Apart from Irrigation Field Laboratory of this College, agricultural land available in the Campus is being effectively utilized for field experiments pertaining to department. Very good Workshop is available with ultra modern facilities in the College to cater the needs of the Department as and when required like fabrication of water measuring devices, etc.



Virtual class room



Class room

Different equipment available in the laboratories are shown below



Digital Water Level Recorder



Total Station





Soil Moisture Probe **Digital Planimeter**



Polyhouses for conduction various Ug, Pg, PhD research works





Linear (Horizontal) NFT Hydroponics units




Vertical NFT Hydroponics units and Ebb and Flow Hydroponics system




Deep water culture Hydroponics system and Nutri-grow towers aeroponics system







Aeroponics systems developed by Dr. N.T.R College of Agricultural Engineering, Bapatla





Dutch Bucket Hydronics system



Grow bag system with drip irrigation





Irrigation Field Laboratory

6.4.5. Conduct of Practical and Hands-on Training

Practical classes for the Ph.D. programme in Soil and Water Conservation Engineering are conducted regularly in the respective laboratories as per the approved class timetable. The Department ensures effective practical exposure to students through well-planned laboratory sessions utilizing the available instruments, equipment, machinery, tools, and experimental facilities. The practical curriculum is designed to complement theoretical instruction and to strengthen the analytical, technical, and research capabilities of the students. The laboratories are adequately equipped to provide hands-on training in various areas of Soil and Water Conservation Engineering, including Irrigation systems, agricultural field irrigation lab, soil dynamics, Hydroponics, Hydraulic lab, instrumentation, and model watershed. Faculty members and technical staff supervise the practical sessions to ensure proper utilization of laboratory resources and to facilitate experiential learning among students.

Special emphasis is given to developing practical competence, problem-solving ability, and research aptitude through laboratory experiments, demonstrations, field observations, equipment handling, calibration exercises, and data analysis. Students are encouraged to actively participate in experimental work, field irrigation, drip and sprinkler testing and evaluation, and research-oriented activities relevant to current challenges in Irrigation, Soil and Water Conservation engineering. As part of experiential learning and exposure to advanced technologies, students are deputed to reputed ICAR research institutes, Central Government organizations, and industries for a period of one month to undergo hands-on training during the Second Year First Semester. These training programmes provide valuable exposure to modern research facilities, advanced instrumentation, industrial practices, and emerging technologies in agricultural engineering.

Practical classes are conducted in respective laboratories as per class Time-Table with the instruments and equipment, etc. available. During Second Year First Semester, students are sent to ICAR research institutes (CRIDA, Hyderabad; CIAE, Bhopal; CIAE, Regional Centre, Coimbatore, etc.), Central Government Organizations (NRSC, Kolkata and Hyderabad, etc.), State Government Organizations (APSAC, Vijayawada etc.) and industries for one month to impart hands-on training to them. PG students are trained on various computer software, namely, AutoCAD, ProE, SURFER, CRIWAR, CROPWAT, SURDEV, MODFLOW, DRAINMOD, SALTMOD, WINFLUME, HYDRUS, RS & GIS, etc. during practical classes. Educational videos are played to the students for better understanding of the subject in various courses.

Industrial/Organizational exposure to students

S.No.	Admission Batch	Organization	Period
1	2020-21	Roots Irrigation Pvt. Limited, 55-F, Vilankurichi Road, Aringar Anna Private Industrial Estae, Coimbatore – 641 035	1st October, 2021 to 31st October, 2021
	2020-21	Nano Scientific Research Centre #604, Siri Estates, Nagarjuna Nagar Yellareddy Guda Ameerpet, Hyderabad - 500073	1st October, 2021 to 31st October, 2021
2	2021-22	WALAMTARI, Himayatsagar, Rajendranagar, Hyderabad - 500030	14th November, 2022 to 13th December, 2022
3	2022-23	Central Institute of Tool Design, Balanagar, Hyderabad - 500037	6th December, 2023 to 5th January, 2024
4	2023-24	IHG farms Pvt. Ltd., Hyderabad	1st October, 2024 to 31st October, 2024
5	2024-25	Sense Semiconductor & IT Solutions PVT.Ltd. 504, North Block, Lakshmi's Sree Likitha Pride, Mangalagiri – 522 503	15th September, 2025 to 14th October, 2025

6.4.6. Supervision of Students in Ph.D. Programmes

Each Ph.D. scholar shall be assigned a faculty member from the Department of Soil and Water Conservation Engineering by the Head of the Department, in consultation with the University Head of the Department, to supervise the scholar's course work and research activities. The assigned faculty member shall serve as the Chairperson of the Student Advisory Committee. In addition, three Advisory Members shall be appointed by the Dean of Post Graduate Studies based on the recommendation of the Chairperson through the Head of the Department. Of these, one member shall be from major department, while the other two shall be from a minor and supporting departments under Agricultural Engineering or Agriculture, depending on the scholar's academic and research requirements. The Student Advisory Committee shall guide the scholar in the selection of courses in the major and minor fields, identification and planning of the research problem, and all other academic and research-related matters. Ph.D. scholars shall complete the prescribed course work during the initial two semesters of the programme and subsequently undertake intensive research work leading to the preparation of the doctoral thesis. The progress of each scholar's research work

shall be reviewed periodically i.e., once in a month by the Student Advisory Committee along with the Head of the Department to provide necessary guidance and recommendations.

During third semester, each scholar shall appear for a comprehensive viva-voce examination conducted by an external Professor in Soil and Water Conservation Engineering from another State Agricultural University or ICAR Institute nominated by the Dean of Post Graduate Studies. The examination shall assess the scholar's proficiency in course work, research methodology, subject knowledge, and progress of doctoral research. The Student Advisory Committee shall supervise and monitor all stages of the research work, including experimentation, data collection and analysis, interpretation of results, publication of research findings, and thesis preparation. Upon completion of the research work, a pre-submission seminar shall be conducted by the Head of the Department in the presence of the Student Advisory Committee and faculty members of the department to obtain suggestions and corrections, if any. After incorporating the recommended corrections, the thesis shall be forwarded to the Dean of Post Graduate Studies through the Head of the Department. The Dean of Post Graduate Studies shall send the thesis to external examiners from other State Agricultural Universities or ICAR Institutes for evaluation. Based on the recommendations of the examiners, the thesis may be accepted with minor or major corrections or rejected for the award of the Ph.D. degree. After incorporating all the corrections suggested by the external examiners, the scholar shall present a final viva-voce examination in the presence of the Student Advisory Committee, external examiner(s), and faculty members of the department to assess the quality and originality of the research work and verify compliance with the recommendations of the examiners. Finally, the corrected thesis shall be submitted to the Head of the Department and the Dean of Post Graduate Studies for the award of the Ph.D. degree. The details of Ph.D. students who were supervised and awarded degree by the faculty under Dept. of Soil and Water Conservation Engineering during last five years is given below.

Guidance by Faculty in the department of Soil and Water Conservation Engineering Ph. D. (Agril. Engg.)

S. No	I.D.No	Name	Chairman	Title of the Study
1	BEP/2020-05	Mr. Ch. Apparao	Dr. A. Mani Professor & University Head Dept. of Soil and Water Engineering Dr. NTR CAE, Bapatla	**2020 Admitted Batch Students**
2	BEP/2020-06	Ms. P. Indraja	Dr. H.V. Hema Kumar Professor	Explicit Spatial Assessment of Climate Change Impact on Crop

			Dept. of Soil and Water Engineering Dr. NTR CAE, Bapatla	Production using Crop Modelling and Geo Spatial Technology in Visakhapatnam Region of Andhra Pradesh
2021 Admitted Batch Student				
1	BEP/2021-01	Er. Nandeppa gari Subhasree	Dr. B. Krishna Rao Principal Scientist & Head ICAR-IISWC, RC, Ballari, Karnataka	Development of Sensor Based Real Time Soil Moisture and Flow Measurement and Regulation System for Increasing Water Use Efficiency
2022 Admitted Batch Student				
1	BEP/2022-01	Mr. R. Rana Prathap	Dr. G. Ravi Babu Professor & Head Dept. of Soil and Water Conservation Engineering Dr. N.T.R CAE, Bapatla	Optimization of Water-Energy nexus and its linkage to crop water productivity in the Kommamuru canal command area

6.4.7. Feedback of Stakeholders (Students, Parents, Industries, Employers, Farmers, etc.)

Students are very happy with the conduct of the Ph.D. (Agricultural Engineering) academic programme being handled at Dr N.T.R. College of Agricultural Engineering, Bapatla.

Feedback of Ph.D. (Agril Engg) Passed Out Students

I, R. Sudha Rani, BEP/2019-06 had compled my Ph.D in the Department of Soil and water Engineering at Dr. NTR College of Agricultural Engineering, Bapatla. from 2019-2022. Iam pleased to Express my gratitude towards my teachers who imparted me knowledge along with course curriculum. Support and guidance I received durring Progress of my project work helped a lot in timely completion of my research. I always had a good learning atmosphere in the classrooms and also with the teachers. I learned the discipline, behaviour Ethics & much more from my proficient teachers who played a key role in Self turning my career as an Teaching Associate in BJR Agricultural college, Srcilla after my Ph.D degree

Yours faithfully,
R. Sudha Rani
12/05/2026.

I, Er. Ch. Apparao, Bearing ID No: BEP-20-05, had completed my Ph.D. in Soil and Water Conservation Engineering at the College of Agricultural Engineering, Bapatla.

I am pleased to express my sincere gratitude towards all the professors and teachers who imparted valuable knowledge throughout my doctoral programme. The constant support, encouragement, and guidance extended during the course work and research period greatly helped me in the journey of my Ph.D. research.

The department provided an excellent academic environment with highly experienced faculty members, well-equipped laboratories, and a positive atmosphere for learning and research. I gained not only technical expertise in Soil and Water Conservation Engineering but also professional ethics, discipline, research aptitude, and confidence from my respected teachers.

The encouragement and mentorship provided by the faculty played a significant role in shaping my professional career and research capabilities. I shall always remain thankful to the department and faculty members for their valuable support and guidance throughout my Ph.D. journey and self-tuning my career as Scientist (Agril.Engg.) at RARS, Maruteru.

Yours faithfully,

Ch. Apparao.

Scientist

(Agril.Engg..)

RARS,

Maruteru

Feedback of External Experts

भाकृअनुप- भारतीय मृदा एवं जल संरक्षण संस्थान, अनुसंधान केंद्र,
छावनी, बल्लारी -583 104, कर्नाटक, भारत
ICAR- Indian Institute of Soil & Water Conservation, Research Centre,
Cantonment, Ballari-583 104, Karnataka, India
Office: 08392-242164, (Res) 08392-242534, Mobile:9574715074
Email: soilcons21bly@gmail.com, b_krishnarao@rediffmail.com

TO WHOMSOEVER IT MAY CONCERN

The students from Dr. NTR College of Agricultural Engineering, earlier known as College of Agricultural Engineering, Bapatla, have undergone In-plant training at ICAR-Indian Institute of Soil and Water Conservation Research Centre since its establishment. During the training period, the students were exposed to various watershed management practices including soil erosion control measures, rainwater harvesting techniques, contour bunding, check dam and farm pond concepts and integrated watershed management approaches. They also learned the planning, design and execution of various soil and water conservation structures and preparation of watershed detailed project reports (DPRs).The students from the above-mentioned college were very punctual and sincere throughout the training period. They actively participated in field visits, demonstrations, technical discussions and ongoing research activities related to watershed management and conservation of soil and water resources and showed dedication towards the completion of the tasks assigned to them.

Head of the Centre
केंद्र के प्रमुख / Head of the Centre
भाकृअप - भा.मृ-ज.सं.सं, अ.कें, बेल्लारी
ICAR-IISWC, RC, BALLARI

DEPARTMENT OF SOIL AND WATER CONSERVATION ENGINEERING
College of Agricultural Engineering and Technology
Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth,
Dapoli-415 712, Dist: Ratnagiri (M.S)
Phone No. 02358/282414 Mobile No. 9422382074 email blayare1967@gmail.com

Dr. B. L. Ayare, M.Tech.,Ph.D.(SWCE)
Professor & Head

No. CAET/SWCE/ 49 /2026
Date : 15/05/2026

TO WHOM SOEVER IT MAY CONCERN

As the Head of the department of the Soil and water conservation Engineering, College & Agril. Engineering and Technology, Dapoli, Dist- Ratnagiri (Maharashtra), I have attended the examination duty and worked as External Examiner for conducting comprehensive oral examinations of M.Tech (SWCE) Students during the year 2024-25 at NTR, College of Agril. Engineering, Baptala. I also guided the third Year B.Tech. Students. The students hard working, sincere and punctual in their work. The College Campus is also good, and well developed.

Dr B. L. Ayase
(Head, SWCE)
Professor and Head
Department of Soil and
Water Conservation Engineering,
College of Agril. Engg. and Tech. Dapoli

Feedback of Training Institutes/Organizations

IHG FARMS PRIVATE LIMITED

4-69-43/A, Laxmi Gangaga Enclave, Pipeline Road, Jeedimetla, Hyderabad, Telangana – 500055

TO WHOM EVER SO IT MAY CONCERN

We are highly impressed with the hydroponics unit established at "Dr. NTR College of Agricultural Engineering", "Bapatla, Andhra Pradesh, India". The unit is being maintained effectively with proper nutrient management and operational practices. The students have shown excellent interest and involvement in learning hydroponic cultivation and modern agricultural technologies during the training program in campus and also at our organization. The facility provides valuable practical exposure and promotes innovation, skill development, and entrepreneurship among students. We appreciate the efforts in promoting advanced hydroponic systems and wish them continued success.

Satyareddy

(Satyanarayana Reddy)
Managing Director
IHG Farms Pvt. Ltd.

राष्ट्रीय वनस्पति स्वास्थ्य प्रबंधन संस्थान
National Institute of Plant Health Management
कृषि, सहकारिता एवं किसान कल्याण विभाग
Department of Agriculture, Cooperation & Farmers Welfare
कृषि, सहकारिता एवं किसान कल्याण मंत्रालय, भारत सरकार
Ministry of Agriculture & Farmers Welfare, Government of India

रा व स्वा प्र सं
N I P H M

Tele -Fax : 040-24015374
E-mail : niphm@nic.in
http://niphm.gov.in

Rajendranagar
Hyderabad - 500 030.
Telanagana, INDIA

TO WHOMSOEVER IT MAY CONCERN

National Institute of Plant Health Management, Hyderabad started the In-plant training since 2014 for the third year students from Dr. NTR College of Agricultural Engineering, earlier known as College of Agricultural Engineering, Bapatla. The students from the above mentioned college are very punctual and sincere during their training period at NIPHM. The students were involved in the on-going research projects of the institute during the training period and have actively participated in the institute activities and also dedicated towards the completion of the tasks assigned.

udaya Bhanu. M

Dr. M Udaya Bhanu
Scientific Officer (PHE)

Feed back from parents of students

As a parent, Iam very happy with the quality of education and discipline maintained at Dr. NTR college of Agricultural Engineering. The college provides excellent guidance along with the practical exposure in agricultural engineering, hydroponics, Irrigation, and modern farming technologies. Faculty members are Supportive, experienced, and approachable for both Students and parents. The Campus environment is Safe, clean, and Student-friendly, encouraging overall personality development. Various workshops, field visits, and hands-on training programs help Students gain real-world knowledge and Confidence. I appreciate the college management for focusing not only on academics but also on innovation, research, discipline, and Career Opportunities for Students.

Iam very much happy that my daughter D.Deepthi Studying M.Tech in the Department of Soil & water Conservation Engineering at Dr. NTR College of Agricultural Engineering, Bapatla.

Yours Obediently
D. Sulochana

6.4.8. Student Intake and Attrition in the Programme for Last Five Years

Admissions to the Ph.D. program in Soil and Water Conservation Engineering are conducted through a transparent, merit-based selection process. Since the 2018–19 academic year, admissions have been facilitated via the Senior Research Fellowship (SRF) entrance examination, which is organized annually by the Indian Council of Agricultural Research (ICAR). This is followed by an evaluation of the interview performance, while strictly adhering to the reservation policy of the Government of Andhra Pradesh. Before the implementation of the ICAR centralized admission system, Acharya N.G. Ranga Agricultural University (ANGRAU) conducted separate entrance examinations for each specialization based on the prescribed M. Tech. (Soil and Water Conservation Engineering) syllabus. The admission procedure is designed to ensure the selection of academically competent and research-oriented candidates with strong foundational knowledge of agricultural engineering and allied disciplines. The program attracts students from various regions of the country through the ICAR admission system, thereby enhancing academic diversity and promoting an interdisciplinary research culture within the department.

In addition to regular admissions, 15% of the total intake is allocated under the ICAR quota, wherein students are admitted with ICAR fellowship support. This fellowship enables scholars to pursue high-quality research in priority areas such as Soil and Water Conservation, Irrigation and drainage applications, precision farming, automation, and sustainable agricultural production systems. The program also provides opportunities for in-service candidates, with the provision of one seat for regular students in each department and one seat exclusively earmarked for in-service faculty members of ANGRAU. This facilitates the capacity building and professional advancement of university faculty and strengthens institutional research capabilities. The Department has consistently maintained a satisfactory level of student enrolment and research output over the years. During the last five academic years, 11 scholars successfully completed and were awarded Ph.D. degrees in Soil and Water Conservation Engineering. The research conducted by these scholars has contributed significantly to the development of location-specific technologies, Irrigation management practices, energy-efficient systems, Hydroponics and sustainable agricultural engineering solutions relevant to the farming communities.

Student Intake and Attrition in Ph. D (Soil and Water Engineering)

Name of the Degree Programme	Actual students admitted in last five years						Attrition (%)					
M. Tech (Ag.Engg.) Soil and water cons engg	2020 - 21	2021 - 22	2022 - 23	2023 - 24	2024 - 25	2025 - 26	2020 - 21	2021 - 22	2022 - 23	2023 - 24	2024 - 25	2025 - 26
	3	1	1	1	-	-	33.33	0	0	100	-	-

6.4.9. ICT Application in Curricula Delivery

The department has smart classroom, interactive board, audio visual facilities, etc. for teaching and for conducting seminars and progress of the students' projects. All the teachers use LCD projector and deliver the curriculum through Power Point Presentation. Also, there is one dedicated classroom with good lighting and ventilation being used for Ph.D. (Agricultural Engineering) degree programme in the department of Soil and Water Engineering for conducting theory classes. The classroom is equipped with LCD projector, chalk board, furniture, etc.

College has Virtual Classroom established under NAHEP-IDP with all the required equipment to connect all the campuses and anywhere in the world. Students are encouraged to deliver presentations through power point presentations.

Training programmes/Short course/Workshops/Conferences Organized

Year	Title of the Training programme/ Short course/ Workshop/Conference	Period		Venue	Organized by	Source of Funding	No. of Participants
		From	To				
2023	Training on "Hydroponics-an era of modern agriculture "	08th November	11th November	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	IDP	86 students from 08 ICAR accredited colleges of ANGRAU

2024	Training on “ Cultivation of crops through hydroponics’	12th October		Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ANGRAU	36 farmers
2025	Field Demonstration Cum Training on “Hydroponics Cultivation Methods “	08th April	08th April	Stuvartupuram	Dr N.T.R CAE, Bapatla	ICAR (TSP)	53 farmers
2025	Field Demonstration Cum Training on “Hydroponics Cultivation Methods “	09th April	09th April	Arumbaka, Cherukupalli	Dr N.T.R CAE, Bapatla	ICAR (TSP)	55 farmers
2025	Training on “Hydroponics Cultivation Methods”	08th Aug	08th Aug	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	39 Farmers
2025	Training and Demonstration on Hydroponic Units	12th Nov	12th Nov	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	22 Farmers
2025	Hands-on Training on Soil-less Cultivation	20th Nov	20th Nov	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	30 Farmers
2025	Training Program on Hydroponics and Aeroponics	25th Nov	25th Nov	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	27 Farmers
2025	Training Program on Hydroponics and Crop Management	12th Dec	12th Dec	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	34 Farmers
2025	Hands on Training Program on Hydroponics	17th Dec	17th Dec	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	28 Farmers
2025	Training on “Maintenance of Hydroponics Units”	18th Dec	18th Dec	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	21 Farmers
2025	Field Demonstration at Jandrapeta on Cultivation of Crops in Hydroponics	23th Dec	23th Dec	Jandrapeta	Dr N.T.R CAE, Bapatla	ICAR (TSP)	33 Farmers
2026	Field Demonstration at Gorantla on different hydroponics and Aeroponics	04th Jan	04th Jan	Gorantla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	32 Farmers

2026	Training on “Hydroponics and aeroponics”	09th Jan	09th Jan	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	25 Farmers
2026	Field Demonstration on Hydroponics Units	11th Jan	11th Jan	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	36 Farmers
2026	Training on “Hydroponics and Soilless Cultivation Systems”	18th February 2026	19th of February 2026	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	12 staff from Krishi Vigyan Kendra’s (KVKs) under ANGRAU
2026	Hands on Training Cum installation of Hydroponic units (Distribution)	27th Feb	27th Feb	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	37 Farmers
2026	Training on “Hydroponics and Soilless Cultivation Systems”	07th Mar	07th Mar	Dr N.T.R CAE, Bapatla	Dr N.T.R CAE, Bapatla	ICAR (TSP)	20 Farmers



Student trainees from 8 ICAR accredited college od ANGRAU



Students undergoing hydroponics training





Staff from KVK's undergoing hydroponics training



Release of Booklet on Hydroponics for distribution to farmers

Farmers trainings at college





Field demonstrations on hydroponics









Total Number of Publications Year-wise

Number of Publications					
Year	**Research Papers**	**Books**	**Training Manuals**	**Conference Papers**	**Total**
2021	03	-	-	02	05
2022	10	-	-	02	12
2023	05	01	01		07
2024	16	-	01		17
2025	06	-	-	03	09
2026	06	-	01	-	07
Total	46	01	03	07	57

List of Research Papers Published by Faculty of Soil and Water Engineering

S. No.	Authors	Year	Title of the paper	Name of the Journal	Vol. and Page Numbers
1	Er B. Chennakesavulu G. Ravi Babu A. Mani D. V. Swami K. Umakrishna	2021	Benefit Cost Ratio as influenced by Different Irrigation and Fertigation Levels on Guava Cv. VNR Bhihi	*International Journal of Genetics*	13: 801-803
2	Er B. Chennakesavulu G. Ravi Babu A. Mani D. V. Swami K. Umakrishna	2021	Effect of Irrigation and Fertigation Scheduling on Vegetative Growth and Yield of Guava cv. VNR Bhihi under High Density Planting System	*International Journal of Current Microbiology and Applied Science*	10(2): 2671-2678
3	Raja Kumar, A. Mani, G. Ravi Babu, M. Martin Luther and CH. Sujani Rao	2021	Effect of Drip Fertigation on Dry Matter Production and Nitrogen Use Efficiency of Maize (Zea mays)	*International Journal of Current Microbiology and Applied Sciences*	10(1): 3212-3222
4	Er Ch. Murali Krishna M. V. Ramana H. V. Hema Kumar B. Ramana Murthy N.V. Sarala Sa Gangadhara Rao	2022	Assessment of Relationships between Rainfall and Canal Water for Maximization of Yield of Major Crops under Telugu Ganga Project command in Andhra Pradesh	*Biological Forum – An International Journal*	14(2): 1377-1388
5	E B. V. Mohana Rao A. Mani B. Sarojini Devi S. Vishnu Vardhan Dr Ch. Sujani Rao	2022	Innovative Trend Analysis (ITA) of Rainfall for Kadiri Watershed Area	*The Andhra Agric. Journal*	69 (2): 303-310
6	Er P. Sambasiva Rao M. V. Ramana K. Madhusudhana	2022	Flood Frequency Analysis of Araniar Medium Irrigation Project in	*International Journal of Environment and*	12(11): 538-544

	Reddy A. Ashok Kumar		Chittoor District by using Gumbel's Distribution	*Climate Change*	
7 Er	S. Janakiramudu K. N. Raja Kumar R. Ganesh Babu Dr Ch. Someswara Rao	2022	Optimization of nutrient concentration and grow media for stevia crop under hydroponic farming	*The Pharma Innovation*	11(12): 5603-5608
8 Er.	N. Ashok Kumar M. V. Ramana M. Raghu Babu B. Ravindra Reddy Dr P. Prasuna Rani	2022	Optimization of Available Water Resources For Sustainable Agriculture in Anantapur District	*International Journal of Agriculture Sciences*	14: 11810-11814
9	Chitte Karishma, B.Krishna Rao, K.Yella Reddy, P.K. Mishra, A. Mani, S.Annapurna, B.Hari Babu V.Srinivas Rao	2022	Low-Cost Plastic Materials for Construction of Flow Measuring Structures	*Biological Forum-An International Journal*	14(4): 267-270,2022
10 Er	V.Tejaswini G. Ravi Babu H. V. Hema Kumar BVS Prasad Ch. Sujani Rao	2022	Status of Water Quality from Agriculture Drains in Guntur District, Andhra Pradesh, India	Current Journal of Applied Science and Technology	41(24): 1-9
11 Er	R. Sudha Rani H. V. Hema Kumar A. Mani B. Sreenivasula Reddy Ch. Sujani Rao	2022	Soilless Cultivation Technique, Hydroponics-A Review	Current Journal of Applied Science and Technology	41(13): 22-30
12 R.	R. Sudha Rani H. V. Hema Kumar A. Mani B. Sreenivasula Reddy Ch. Sujani Rao	2022	Cultivation of Basil under Hydroponics with Nutrient Film Technique in Controlled Environment	Ecology Environment and Conservation	28: 237-242
13	M. Vinayak , G. Vani , N. Subhasree and Ch. Ratna Raju	2022	Effect of Zai Pit Technology on the Infiltration Capacity of the Soil	*International Journal of Environment and Climate Change*	12(12): 999-1005
14 Er	P. Indraja H. V. Hema Kumar A. Mani.	2023	Estimation of Ground Water Recharge in the Command Area of Krishna Central Delta	International Journal of Environment and Climate Change	13: 1317-1324

	K.V. S. Rami Reddy Ch. Sujani Rao				
15 Er	Preeti Sajjan M. V. Ramana K. Krupavathi P. Sumathi	2023	Infrared Thermometry Studies for Estimation of Crop Water Stress Index	*International Journal of Environment and Climate Change*	13: 4285-4294
16	D. Sai Gangadhara Rao, Ch. Murali Krishna, H. V. Hema Kumar, B. Sarojini Devi, L. Edukondulu and V. Srinivasa Rao	2023	Assessment of available water and crop water requirements of major crops in Nagarjuna Sagar Right Canal Command Area, Andhra Pradesh	*International Journal of Plant & Soil Science*	35 (3):1-14
17 B.	V. Mohana Rao A. Mani B. Sarojini Devi S. Vishnu Vardhan Ch. Sujani Rao	2023	Spatial and Temporal Variations of Land Use and Land Cover In Kadiri Watershed Using Rs & Gis Techniques	*International Journal of Plant & Soil Science*	35: 23-31
18	Raja kumar K.N., Mani A, Ravi Babu G. And Sujani Rao Ch	2023	Rainfall Distribution and Crop Water Requirement (Cropwat 8.0) of Maize During Kharif and Rabi Seasons of Coastal Andhra Pradesh	*International Journal of Agriculture Sciences*	15 (9):12616-12618
19 Er	P. Indraja H. V. Hema Kumar A. Mani. K.V. S. Rami Reddy Ch. Sujani Rao	2024	Estimation of Monthly Irrigation Water Requirement of Crops Using Cropwat	Journal of Scientific Research and Reports	30: 344-351
20 E	P. Anila M. Raghu Babu K. V. Ramana H. V. Hema Kumar	2024	Development of Sustainable Water Resourcrs Management Plan for Pothakamuru Watershed using RS and GIS	*International Journal of Science and Research (IJSR)*	13: 2319-7064
21	D. Sai Gangadhara Rao, Ch. Murali Krishna, H. V. Hema Kumar, B. Sarojini Devi, L. Edukondulu and V. Srinivasa Rao	2024	Development of Location Specific Decision Support System (DSS) for Adopting the Best Water Management Strategies to Maximize the Production for Nagarjuna Sagar Right Canal Command Area	*International Journal of Plant & Soil Science*	36 (10):400-412

22 Er	K.V.L. Supraja K. Krupavathi G. Ravi Babu Ch. Someswara Rao	2024	Forecasting the polyhouse atmosphere: Assessment of key modeling instruments	*Current Advances in Agricultural Sciences*	16(2): 55-62
23 Er	K.V.L. Supraja K. Krupavathi G. Ravi Babu Ch. Someswara Rao	2024	Demystifying Polyhouse Microclimates: A Review of Modeling Tools and Strategies	*International Journal of Environment and Climate Change*	14: 150-165
24 Er	K.V.L. Supraja K. Krupavathi G. Ravi Babu Ch. Someswara Rao	2024	Spatial Variation of Microclimatic Parameters Inside a Polyhouse with Fog Cooling System	*Journal of Scientific Research and Reports*	30: 643-663
25 K.	N. Raja Kumar, A. Mani, G. Ravi Babu, Ch. Sujani Rao	2024	Influence of Drip Fertigation on Maize Growth Parameters	*Environment and Ecology*	42 (2B) : 881-890
26 K.	N. Raja Kumar, A. Mani, G. Ravi Babu and CH. Sujani Rao	2024	Spatial and Temporal Distribution of Nitrogen in Maize under Drip Fertigation	*International Journal of Plant & Soil Science*	36 (5):375-392
27 Er Sa	Ch. Murali Krishna M. V. Ramana H. V. Hema Kumar B. Ramana Murthy N.V. Sarala Gangadhara Rao	2024	Impact of Telugu Ganga Project on Irrigation Efficiency and Crop Yield in Nellore District, Andhra Pradesh, India	*International Journal of Plant & Soil Science*	36: 451-465
28	Pavithra VK, G Ravi Babu and M Latha	2024	Effect of deficit irrigation on plant growth parameter of sweet corn crop under drip irrigation	*International Journal of Research in Agronomy*	7(12): 523-529
29	V.K.Pavithra, R.GaneshBabu, G.RaviBabu and M.Latha	*2024*	*Estimation of Crop Water Requirement of Sweet corn crop using CROPWAT 8.0 model*	*Madras Agricultural Journal*	*111:4-6*
30	V.K.Pavithra, R.GaneshBabu, G.RaviBabu and M.Latha	2024	Simulation of Soil Water Dynamics using the Hydrus-1d Model for Sweet Corn under Drip Irrigation	*Madras Agricultural Journal*	113:1-3

31	Rana Prathap, K. Krupavathi, K. Madhusudhan Reddy, A. Ashok Kumar	2024	Design and Development of Automated Nutrient Regulation System for Hydroponics Unit	*Environment and Ecology*	42 (3A): 1105-1112
32	Rana Prathap R., Ratna Raju Ch.	2024	Estimation of Crop Water Requirement and Determination of Irrigation Schedule for Tomato Crop by using CROPWAT 8.0 Model	*Environment and Ecology*	42 (1A) : 341-346
33 Er	N. Naga Hari Sairam A. Mani H. V. Hema Kumar K.V. S. Rami Reddy K. Srinivasulu	2024	Monthly actual evapotranspiration estimation using Google Earth engine in the Godavari districts of Andhra Pradesh	International Journal of Research in Agronomy	1: 203-209
34 Er	R. Sudha Rani H. V. Hema Kumar A. Mani B. Sreenivasula Reddy Ch. Sujani Rao	2024	Analysis of the Environmental Parameter of Hydroponic Crop Grown under Polyhouse Conditions in Different Seasons	International Journal of Environment and Climate Change	14: 964-972
35 K.	K.N. Raja Kumar, A. Mani G. Ravi Babu, and CH. Sujani Rao	2025	Assessment of Hydrus Model Accuracy in Simulating Soil Moisture under Drip Irrigation for Maize	*International Journal of Environment and Climate Change*	15 (8): 384-395
36 D.	Sai Gangadhara Rao, Ch. Murali Krishna, H. V. Hema Kumar, B. Sarojini Devi, L. Edukondulu and V. Srinivasa Rao	2025	Assessment of crop water requirements of Nagarjuna Sagar Right Canal Command Area, Andhra Pradesh, India	*International Journal of Environment and climate change*	15 (6):313-333
37 Er	Anandita Rajput G. Ravi Babu K. Krupavathi Dr. B. Raj Kiran	2025	Design and Development of Aerogarden System for Exotic Crop Cultivation	Economics Environmental condition	31: 483-S488
38 Er	R. Sudha Rani H. V. Hema Kumar A. Mani B. Sreenivasula Reddy Ch. Sujani Rao	2025	Evaluation of the Cost Economics of Hydroponic System for the Crop Grown in Polyhouse Conditions	Journal of Experimental Agriculture International	47: 140-145
39	Kishore	*2025*	Sustainable Intensification	*Legume*	*LR-5571 [1-9]*

	Nalabolu, Ch. Ratna Raju, D. Srigiri V. Shobhan Naik, K. Arun Kumar, B.V. Mohana Rao		of Rainfed Groundnut Farming: Assessing Mechanized In situ Rainwater Conservation Practices for Yield, Energy and Carbon Efficiency	*Research-An International Journal*	
40	Vani G, Subhasree N, Vinayak M, Ratna Raju Ch	2025	Comparison of crop water requirement using aqua crop model and pan evaporimeter in zaipit technology and furrow irrigation methods	*International Journal of Ecology and Environmental Sciences*	7(2): 41-47
41	Chitte Karishma, B.Krishna Rao, A. Mani, B.Hari Babu and V.Srinivas Rao	2026	Cost Comparison of Materials for Development of Flow Regulation and Control unit	*International Journal of Agriculture Extension and Social Development*	9(1):101-103
42	Ashokkumar, N., Ramana, M. V., Babu, M. R., Rani, P. P., & Reddy, B. R.	2026	Monthly Rainfall Prediction Using SARIMA models for 63 Mandals of Anantapur District, Andhra Pradesh, India	*International Journal of Environment and Climate Change*	16(5): 21–53.
43	Ashokkumar, N., Ramana, M. V., Babu, M. R., Rani, P. P., & Reddy, B. R	2026	Design of Location Specific Farm Ponds Software	*Asian Journal of Advances in Agricultural Research,*	26(4):1–15.
44	Ashokkumar, N., Ramana, M. V., Babu, M. R., Rani, P. P. and Reddy, B. R	2026	Assessment of Meteorological Drought Dynamics in Anantapuramu District, India	*Journal of Experimental Agriculture International*	48(4):654–673
45	Kishore Nalabolu, Madhusudhan Reddy Karakala, Apparao Chodisetti, Bhaskara Rao Ijjurouthu, Narayanaswamy Gutta, Nataraj Kolavanahalli Chikkamuniyappa, Murali Krishna Chitte, Arun KumarKondeti,	*2026*	Ridge Regression Modeling of Evaporation Reduction Strategies for Small-Scale Water Storage in Semi-Arid Regions	*Agri Engineering*	*8, 55*

	Veera Prasad Godugula, Rajakumar Kommathoti Navaneetha, Mohana RaoBoyinapalli Venkata, Ratnaraju Chebrolu, Srigiri Doppalapudi 5 and Shobhan Naik Vankanavath				
46	Ashokkumar, N., Ramana, M.V., Babu, M.R., Rani, P.P. and Reddy, B.R.	2026	Cluster analysis for 63 mandals in Anantapuuramu district using SPSS	*Internation al Journal of Agriculture and Food Science*	8(4): 76-78

List of Books Published

Sl. No.	Authors	Year	Title of the Book	Name of the Publisher
1	**Ravi Babu, G., Ganesh Babu, R., Krupavathi, K. and Raja Kumar, K.N.**	2017	Irrigation Engineering	VGS Polytechnic Series, VGS Book links, Vijayawada

List of Training Manuals Prepared

Sl. No.	Authors	Year	Title of course
1	K. Krupavathi, G. Ravi Babu, Dr. R. Ganesh Babu, Er. Ch Ratna Raju and Er. M. Karunaya	2023	Hydroponics – An era of Modern Agriculture
2	K. Krupavathi, G. Ravi Babu, Dr. H. V. Hema Kumar, Dr. N. Ashok Kumar, Er. Ch Ratna Raju and Er. M. Karunaya	2024	Hydroponics -Aadunika vyvasaya vidhanam
3	K. Krupavathi, G. Ravi Babu, Dr. K.N. Raja Kumar, Dr. N. Ashok Kumar and Er. M. Karunaya	2026	Hydroponics- A smart and Sustainable Approach to soilless Farming

List of Conference Papers Presented by Faculty

Sl.No.	Name of the Authors	Title	Year	Details of the seminar / conference with place and date
1	K. Krupavathi, R. Sudha Rani, K. N. Rajkumar and H. V. Hema Kumar	Performance evaluation of nutrient film technique for the cultivation of Red Amaranths (Amaranthuscruentus)	2021	5th International Conference on Advances in Agriculture, Environmental and Biosciences for Sustainable Development (AAEBSSD-2021), via zoom video conference App. and held on 05-07th August.
2	K. Krupavathi, R. Sudha Rani, K. N. Rajkumar and H. V. Hema Kumar	Performance evaluation of nutrient film technique for the cultivation of Red Amaranths (Amaranthuscruentus)	2021	5th International Conference on Advances in Agriculture, Environmental and Biosciences for Sustainable Development (AAEBSSD-2021), via zoom video conference App. and held on 05-07th August.
3	R. Sudha Rani, N. Subha Sree, H.V. Hema Kumar and A. Mani	Hydroponics-Soilless Cultivation Techniques- a Review.	2022	Annual post graduate students national conference (APGSNC) on Climate Resilient and Eco-friendly Strategies Towards Sustainable Agriculture and Food Security. At agricultural college, Bapatla and held on 6-7th May.
4	R. Sudha Rani, K. Krupavathi, K.N. Rajkumar. and H.V. Hema Kumar	A Case Study of Cultivation of Red Amaranthus in Hydroponics.	2022	IVth international conference in Hybrid mode on innovative and current advances in agriculture and allied sciences (ICAAS-2022) at Himachal Pradesh University, summer Hill, Shimla (H.P) India and held on 12-14th June.
5	R. Sudha Rani, K. Krupavathi and H. V Hema Kumar	Effect of Base Medias in Growing of the Basil Crop in NFT Hydroponics System.	2025	2nd National Seminar on Emerging Trends in Agriculture and Allied Sciences: A Pathway to Food Security. ENCOURAGE society, Bengalure & AgriTech Publication, Bengaluru and held on 20-21 September. (**Best oral presentation)**
6	R. Sudha Rani, H. V. Hema Kumar, A. Mani,	Analysis of the Environmental Parameters of Hydroponic Crop	2025	11th National Conference: FutureAg-2025. “Advancing Food & Farming for

	Boreddy Sreenivasula Reddy and Ch. Sujani Rao.	Grown under Polyhouse Conditions in Different Seasons.		Sustainable Future". Suresh Gyan Vihar University, Jaipur, Rajasthan and held on 9-10 September
7	R. Sudha Rani, H. V. Hema Kumar, A. Mani, Boreddy Sreenivasula Reddy and Ch.Sujani Rao.	Biometric Parameters of Crops Grown under Different Substrates and Concentration of Nutrient Solution.	2025	11th International Conference on Recent Advances in Agricultural, Biological & Applied Sciences for Eco-Friendly Development (RAABASED-2025). Maya Devi University, 3Dehradun, Uttarakhand, I4ndia and held on July 25-27.

6.5.3. Learning Resources

PG students are trained on various computer software, namely, AutoCAD, ProE, SURFER, CRIWAR, CROPWAT, SURDEV, MODFLOW, DRAINMOD, SALTMOD, WINFLUME, HYDRUS, QGIS and Arc GIS etc. during practical classes. Educational videos are played for better understanding of the subject in various courses.

College Library (Digital)

Library and information services play a vital role in supporting high quality education and learning environment amongst students, faculty members and post-graduate scholars. This College is having a separate block for Library. The Library has a good collection of 12,190 subject matter books, 470 gift books, 56 NAHEP-IDP books and 461 project reports. Apart from these books, there are 97 books and 147 books exclusively for SC and ST students. To facilitate the students writing competitive examinations, library is enriched with GRE, GMAT, TOEFL, CAT, general studies, computer books with CDs and different dictionaries (Telugu to English and English to English), handbooks, annual reports, world atlas, etc. Apart from them, there are a wide variety of Telugu and English daily news papers, popular weekly and monthly magazines like *Rythunestham*, *Annadata*, etc. and competitive magazines. They are being kept for students general reading.



Reading facility



Magazine Section





Digital Computer lab and PG computer lab

The issues and returns of the library books are strictly on library card basis. The College Library is equipped with 2 good Xerox machines and 2 computers along with printers. Apart from them, 32 computers with internet facility is being maintaining under digital library. The entire library is equipped with 09 air conditioners for comfortable study of students. Computers having high end configuration with 40 MBPS Internet speed to utilize the electronic information, namely CeRA, CAB Abstracts, KrishiKosh, EBSCO, Indiastat, CRC Netbase e-books, CAB e-books, Elsevier e-books, CMIE, EPWRF India Times Series, EZ Proxy, DELNET, OPAC, etc. for the improvement of academic and research standards. The library is completely under survailence with 14 no of CC cameras monitoring all the time.

Number of books: 12190

Periodicals: 517

6.5.4. Student Development

College is imparting very good education to the students of M.Tech. (Agricultural Engineering) for their overall academic development.

Average Number of Students in Theory and Practical Classes

Student Strength in Theory and Practical Classes

Sl. No.	Name of the Degree Programme	Batch of Students in Theory Class	Batch of Students in Practical Class
1	Ph.D. (Agricultural Engineering) Soil and Water Engineering	2	2

Admission Process

Registration of the courses is being done in the College during Orientation Session for Ph.D. programmes separately. Students have to fill the registration material and the registration slips will be passed on to the respective courses in-charge. Academic Calendar is scrupulously prepared by the University for Ph.D. programmes separately.

Ph.D. (Agricultural Engineering)

Admission process carried out to the Ph.D. programmes is through entrance examination (SRF) conducted by the ICAR every year (since 2018-19) and interview performance as per the reservation policy of the state. Earlier ANGRAU conducted the entrance examinations in each specialization based on the Ph. D. (Agricultural Engineering) syllabus. Apart from regular admission, over and above, 15% of students secure admission under ICAR quota with ICAR fellowship. There is a provision of one seat for regular students in each Department and one seat altogether for in-service faculty of ANGRAU.

Examinations and Evaluation Process

Examinations for all the courses are being conducted with fool proof and transparency. Most of the courses are having theory as well as practical component. Each credit of the course is evaluated for 50 marks.

Ph.D. (Agricultural Engineering)

Mid-semester examinations are being held after 50% of the semester working days in half the syllabus, whereas, semester final theory examinations are conducted after completion of the semester. Mid-semester examinations and semester final theory examinations are conducted for 30 and 100 marks, respectively. Assignments (including term papers) are evaluated for 20 marks.

Mid-semester examination question papers are set by the courses in-charge itself and conduct the examination. Question paper has two parts (Part-A and Part-B). Part-A contains bits in the form of fill in the blanks, multiple choice questions, matching, formulae, statements, one sentence answers, etc. for 15 marks. Part-B contains 3 questions of essay type, derivations,

problems solving, etc. for 15 marks. After examination is completed, the concerned course in-charge evaluates the answer scripts and show to the students for verification within 10 days.

Practical final examination is being conducted in the laboratory, workshop, computer centre, drawing hall or examination hall depending upon the nature of the course. Practical component is evaluated on 50:50 bases, i.e., 50% of the marks for internal evaluation based on record work, experimentation, slip tests, quizzes, attendance, etc. and the remaining 50% of the marks are for final practical examination. Those students who obtained more than 75% of attendance are allowed to appear for both the practical final and semester final theory examinations.

Question papers for semester final theory examinations are also set by the concerned courses in-charge and sent to the University. Question paper has two parts (Part-A and Part-B). Part-A contains short answer questions for 40 marks. Out of 10 questions, student has to attempt 8 questions. Part-B contains 8 questions of essay type, derivations, problems solving, etc. for 60 marks, but, students has to attempt any 5 questions. All these question papers are moderated by senior faculty at University level. Then, moderated question paper is printed and supply the sealed bundles of the question papers to the concerned Colleges one day before the start of the semester final theory examinations. They are put under lock and key by the Associate Dean of the College. On the day of examination, the concerned question paper bundle is opened 15 min prior to the scheduled start of the examination. Examinations are conducted with the help of course in-charge and departmental staff. The examination is held for 3 hours. After examination, all the answer scripts are sealed and put under custody. All the bundles of the answer scripts are sent to University. There, answer scripts are jumbled/randomized and coded. Course in-charge, who dealt the same course, is assigned the evaluation duty by Dean of Post Graduate Studies. Generally, each teacher is allowed to evaluate about 30 answer scripts in a day. After evaluation is completed, the answer scripts are decoded by University and the results are sent to the College. The course in-charge prepares the Performance Register as per the weightage and declares the results. Those students who obtained more than 75% of attendance are permitted to appear for the semester final theory examination.

Students have to score 50% of marks to pass both semester theory examination and practical

final examination. Also, they need to obtain 6.0 Grade Point to clear the course.

Comprehensive written and oral examinations are conducted after the course work is completed. Written examination in both core and minor courses separately is being conducted for Ph.D. students. Ph.D. students have to obtain 70% marksin these written examinations. Oral examination for these students is being conducted by inviting External Examiner from other SAUs or IITs.

Personality Development

College is continuously striving hard to overall development of all the students. Several personality development classes are being conducted regularly apart. Special classes also being conducted for backlog students and SC-ST students. Much emphasis also being given to improving the spoken English in order to perform better in placement interviews.

Research Contingency

ANGRAU is in its budget allocation, PG research contingency fund is allocated annually to support the M.Tech. and Ph.D. students in the department of Soil and Water Conservation Engineering (SWCE).

Department-wise Research Grant during Last Five Years

Year	SWE
2020-21	5,04,000
2021-22	2,07,000
2022-23	2,66,000
2023-24	66,000
2024-25	15,000
2025-26	60,000

Outcome/Output

Patent

Received patent on "**Polymer-pully driven centrifugal pump"** Dr. A. Rama Rao, Dr. G. Ravi Babu, team and Acharya N. G. Ranga Agricultural University

Research Highlights of Ph.D. students

Rathnala Sudha Rani optimized the micro-environment for basil cultivation under hydroponic conditions using Nutrient Film Technique (NFT). The study evaluated different nutrient concentrations

and growing media across four seasons under split plot design. Environmental parameters such as temperature, relative humidity, and CO_2 were analyzed for their influence on crop yield. Statistical optimization using Design Expert software identified suitable nutrient-media combinations for maximum yield during different seasons, with rock wool and vermiculite performing effectively. The research established significant relationships between environmental variables and crop productivity. Economic analysis indicated profitable basil cultivation under hydroponics with potential for further expansion and improved benefit-cost ratio through enhanced polyhouse utilization.

Vallu Tejaswini developed and evaluated a low-cost filtration mechanism for treatment and reuse of wastewater from agriculture and aquaculture sectors. Water quality assessment conducted in Guntur and Prakasam districts revealed excessive salts and pollutants in wastewater sources. A horizontal filtration system using gravel, sand, coir, ceramic rings, wood chips, and activated carbon was designed and tested with different media combinations and proportions. The developed filters significantly reduced chemical pollutants and improved water quality suitable for reuse. Statistical analyses confirmed effective pollutant removal efficiencies, while benefit-cost analysis demonstrated economic feasibility. The developed technology offers an affordable solution for sustainable wastewater management and reuse in agriculture.

B. V. Mohana Rao investigated hydro-geomorphological changes caused by land use dynamics in the Kadiri watershed of Anantapuramu district using GIS, Remote Sensing, and hydrological modelling techniques. Rainfall trend analysis, morphometric assessment, and thematic mapping were carried out to evaluate watershed characteristics and changes over time. The QSWAT model was applied to simulate runoff, stream flow, evapotranspiration, groundwater contribution, and sediment yield from 1982 to 2020. The study revealed significant improvements in agricultural area, plantations, and water bodies following watershed interventions, along with reduced soil erosion and enhanced groundwater contribution. Groundwater potential zones were identified to support sustainable water resource management and long-term agricultural development.

Chitte Karishma developed and evaluated baffle sluice modules to improve water regulation and enhance water use efficiency in canal command areas. The study focused on designing affordable and farmer-friendly flow control units capable of delivering near-constant discharge under varying head conditions. Modules of 7.5, 10, and 15 L/s were fabricated using polypropylene sheets and tested under field conditions in Telangana command areas. Results showed discharge deviations below 10%,

demonstrating reliable performance. The modules significantly improved irrigation efficiency, increasing water use efficiency by 61%, saving up to 45% irrigation water, and enhancing crop yield by 11–14%, thereby supporting sustainable irrigation management.

Nethi Naga Hari Sairam conducted flood inundation modelling in the Godavari districts of Andhra Pradesh using HEC-HMS and HEC-RAS models to assess flood impacts on agriculture. The research integrated hydrological, hydraulic, GIS, and remote sensing techniques to simulate flood depth, velocity, inundation extent, and crop damage. Flood risk assessment revealed severe inundation in vulnerable mandals, causing major paddy crop losses and economic damage during the 2013 floods. The study also identified suitable locations for flood storage ponds and emphasized flood zoning and drainage improvement as mitigation strategies. The developed modelling framework supports real-time flood forecasting and sustainable flood risk management in ungauged river basins.

Pulugu Indraja conducted conjunctive use planning of surface and groundwater resources in the Krishna Central Delta using integrated hydrological and optimization models. SWAT, MODFLOW, and MT3DMS models were applied to simulate runoff, groundwater dynamics, and salinity distribution under different management scenarios. The study revealed that increased groundwater pumping significantly raised salinity levels, while enhanced recharge minimized deterioration in water quality. Linear programming optimization using LINGO identified optimal combinations of canal and groundwater use that maximized net economic benefits under different crop allocations. The research demonstrated that conjunctive management of water resources improves irrigation sustainability, enhances economic returns, and supports efficient crop planning in canal command areas.

Awards and Honors of the Faculty during Last Five Years

Sl. No.	Name	Awards and Honors
1	Dr. G. Ravi Babu	• **State Best Teacher Award** from Government of Andhra Pradesh, India on Teachers Day **2023** • Received **patent on "Polymer-pully driven centrifugal pump" Dr. A. Rama Rao, Dr. G. Ravi Babu, team and Acharya N. G. Ranga Agricultural University**
2	Dr K Krupavathi	• **Meritorious Teacher award** for the year 2023 during awarded during 54th Annual Convocation of Acharya N. G. Ranga Agricultural University • **Outstanding Women researcher** for the year 2025 during

		international conference conducted by Venus International Foundation • **Project of the year award** during **Student project Programme** 2020 (SPP) for the Best Ph.D work from Institute of Scholars presented during 2021 • **Best oral presentation in** 2nd National Seminar on Emerging Trends in Agriculture and Allied Sciences: A Pathway to Food Security. ENCOURAGE society, Bengalure & AgriTech Publication, Bengaluru and held on 20-21 September, 2025 for the paper "Effect of Base Medias in Growing of the Basil Crop in NFT Hydroponics System"

Externally funded projects of Faculty

1. **Dr K. Krupavathi, Principal Investigator for project** entitled "Empowering Agro-Entrepreneurs on Modern Soilless Agriculture through Skill Development for Job Creation and Unemployment Alleviation" sanctioned under (ICAR-TSP). All structures related to hydroponics are established and training was given to about 500 farmers.

6.4.11 Certificate

I, the Associate Dean of the Dr N.T.R College of Agricultural Engineering, Bapatla, Andhra Pradesh hereby certify that the information contained in Sections 6.4 and 6.5.1 to 6.5.7.4 are furnished as per the records available in the college and degree awarding university.

ASSOCIATE DEAN
Dr. NTR College of Agricultural engineering
BAPATLA-522 101